

## Artificial Intelligence (CS-203)

### Student: Solving Mathematical Equations

#### ➤ Module 2. Solving the list of equations

Repeat the following steps to solve equations:

1. Select equation with one variable
2. Isolate the variable on left hand side of the equation
3. Evaluate the right side of equation to get value of the variable
4. Replace the variable with its value in all equation

#### ➤ Code of Solve

```
def solve(eqs, solved=[]):  
    # print (solved)  
    # print (eqs)  
    if len(eqs) == 0:  
        return solved  
    ans = find_one_unknown(eqs)  
    var = ans[0]  
    eq = ans[1]  
    if eq != [] and var is not None:  
        sol=solve_equation(var,eq)  
        value = sol[1]  
        eqs.remove(eq)  
        eqs = replace_item(value, var,eqs)  
        solved.append([var, '=', value])  
        temp= solve(eqs,solved)  
        return temp  
    return solved
```

#### ➤ Logic of isolate

1. Equal lhs unknown then Stop
2. If unknown in rhs then swap sides and recursive call
3.  $X + A = B \rightarrow X = B - A$   
 $X - A = B \rightarrow X = B + A$   
 $X * A = B \rightarrow X = B / A$   
 $X / A = B \rightarrow X = B * A$
4.  $A + X = B \rightarrow X = B - A$   
 $A * X = B \rightarrow X = B / A$
5.  $A - X = B \rightarrow X = A - B$

$$A / X = B \rightarrow X = A / B$$

### ➤ Code of isolate

```
def isolate(var, eq):
    lhs= eq[0]
    op = eq[1]
    rhs= eq[2]
    if var == lhs: #CASE 1: X = A
        return eq
    elif var == rhs:
        return [rhs, op, lhs]
    elif type(lhs) is str:
        return isolate(var, [rhs,op,lhs])
    elif type(rhs) is list and op == '=' and in_exp(var,rhs):
        return isolate(var, [rhs,op,lhs])
    elif in_exp(var,lhs[0]):
        return isolate(var, [lhs[0], '=', [rhs, reverse_op(lhs[1]), lhs[2]]])
    elif lhs[1] == '+' or lhs[1] == '*':
        return isolate(var, [lhs[2], '=', [rhs, reverse_op(lhs[1]), lhs[0]]])
    else:
        return isolate(var, [lhs[2], '=', [lhs[0], lhs[1], rhs]])
```

### ➤ Sample Calls of Isolate

```
>>> e1 = ['A', '=', ['2', '+', ['2', '*', '3']]]
>>> e1
['A', '=', ['2', '+', ['2', '*', '3']]]
>>> isolate('A', e1)
['A', '=', ['2', '+', ['2', '*', '3']]]
>>> e2 = [['2', '+', ['2', '*', '3']], '=', 'A']
>>> isolate('A', e2)
['A', '=', ['2', '+', ['2', '*', '3']]]
>>> e3 = ['4', '=', ['2', '+', ['A', '*', '3']]]
>>> isolate('A', e3)
['A', '=', [['4', '-', '2'], '/', '3']]
>>> e4 = [['2', '+', ['2', '/', 'A']], '=', '4']
>>> isolate('A', e4)
['A', '=', ['2', '/', ['4', '-', '2']]]
>>> e5 = ['4', '=', ['2', '/', ['A', '*', '3']]]
>>> isolate('A', e5)
['A', '=', [['2', '/', '4'], '/', '3']]
>>> e6 = [['2', '-', ['2', '*', 'A']], '=', '4']
>>> isolate('A', e6)
['A', '=', [['2', '-', '4'], '/', '2']]
>>> e7 = ['4', '=', ['2', '/', ['A', '/', '3']]]
>>> isolate('A', e7)
['A', '=', [['2', '/', '4'], '/', '3']]
>>> e8 = [['2', '/', ['2', '/', 'A']], '=', '4']
>>> isolate('A', e8)
['A', '=', ['2', '/', ['2', '/', '4']]]
>>> e9 = [['4', '+', ['5', '+, '3']], '=', ['4', '+, 'A']]
>>> isolate('A', e9)
```

```

['A', '=', [[['4', '+', ['5', '+', '3']], '-', '4']]
>>> e10 = [[['4', '+', ['A', '/', '3']], '=', ['4', '-', '3']]
>>> isolate('A', e9)
['A', '=', [[['4', '+', ['5', '+', '3']], '-', '4']]
>>> e10 = [[['4', '+', ['A', '/', '3']], '=', ['8', '-', '3']]
>>> isolate('A', e10)
['A', '=', [[['8', '-', '3'], '-', '4'], '/', '3']]
>>> e11 = [[['4', '-', ['5', '-', 'A']], '=', ['8', '+', '3']]
>>> isolate('A', e11)
['A', '=', ['5', '-', ['4', '-', ['8', '+', '3']]]]

```

### Steps to solve equations of Example 3

#### Solve.1

$(x + y) = (z - 5),$  (To-find =  $(z * z)$ ),  
 $(6 - x) = 4, (8 / y) = 1/4$   
 1. selection (equation with one unknown)  $(6 - x) = 4$   
 2. isolation  $\rightarrow (x = (6 - 4))$   
 3. solve-arithmetic  $\rightarrow (x = 2)$   
 4. substitution and recursive call

#### Solve 2

$(x + y) = (z - 5),$   $(x = 2)$   
 $(\text{To-find} = (z * z)),$   
 $(8 / y) = 1/4$   
 1. selection (equation with one unknown)  $((8 / y) = 1/4)$   
 2. isolation  $\rightarrow (y = 8 / 1/4)$   
 3. solve-arithmetic  $\rightarrow (y = 2)$   
 4. substitution and recursive call

#### Solve 3

$(x + y) = (z - 5),$   $(x = 2)$   
 $\text{To-find} = (z * z),$   $(y = 2)$   
 1. selection (equation with one unknown)  $((2 + 2) = (z - 5))$   
 2. isolation  $\rightarrow ((z - 5) = (2 + 2)) \rightarrow (z = ((2 + 2) + 5))$   
 3. solve-arithmetic  $\rightarrow (z = 9)$   
 4. substitution and recursive call

#### Solve 4

$(\text{To-find} = (z * z))$   $(x = 2), (y = 2)$   
 $(z = 9)$   
 1. selection (equation with one unknown)  $(\text{To-find} = (9 * 9))$   
 2. isolation  $\rightarrow (\text{To-find} = (9 * 9))$

3. solve-arithmetic → (To-find = 81)

4. substitution and recursive call

### **Solve 5**

[ ] (x = 2), (y = 2), (To-find = 81), (z = 9)

#### **Solve 5.4.3.2.1 Return value:**

[(x = 2), (y = 2), (To-find = 81), (z = 9)]

Print-equations function will display the solved equations in on output window.

## **EXAMPLE 2: TRACING OF COMPLETE STUDENT PROCESS**

**Enter Statement:** Sum of theory and practical marks obtained by a student makes the total marks of the student for a course. Similarly, Theory marks are calculated by adding Internal marks and marks obtained during exams. Moreover, marks of exams can be further distributed into Mid exam and Final Exam. Internal marks can be divided into Quizzes and Assignments. Find total marks obtained by Asif if he has obtained . 8 Marks in Quizzes. 6 marks in Assignments. 10 marks in Mid . 20 marks in Final exam. Finally, 12 in practical lab work.

### **A. TRACING OF TRANSLATE\_TO\_EXPRESSION MODULE**

Input: " sum theory and practical marks obtained by student makes total marks student for course "

Rule: " ?\*a makes ?\*b " , [ ?a = ?b ]

Binding: {'a': ['sum', 'theory', 'and', 'practical', 'marks', 'obtained', 'by', 'student'], 'b': ['total', 'marks', 'student', 'for', 'course']}

Input: " sum theory and practical marks obtained by student "

Rule: " sum ?\*a and ?\*b " , [ ?a + ?b ]

Binding: {'a': ['theory'], 'b': ['practical', 'marks', 'obtained', 'by', 'student']}

Input: " theory "

Input: " practical marks obtained by student "

Output: ['theory', '+', 'practical']

Input: " total marks student for course "

Output: [['theory', '+', 'practical'], '=', 'total']

Input: " theory marks are calculated by adding internal marks and marks obtained during exams "

Rule: " ?\*a calculated by ?\*b " , [ ?a = ?b ]

Binding: {'a': ['theory', 'marks', 'are'], 'b': ['adding', 'internal', 'marks', 'and', 'marks', 'obtained', 'during', 'exams']}

Input: " theory marks are "

Input: " adding internal marks and marks obtained during exams "

Rule: " adding ?\*a and ?\*b " , [ ?a + ?b ]

Binding: {'a': ['internal', 'marks'], 'b': ['marks', 'obtained', 'during', 'exams']}

Input: " internal marks "

Input: " marks obtained during exams "

Output: ['internal', '+', 'marks']

Output: ['theory', '=', ['internal', '+', 'marks']]

Input: " marks exams can be distributed into mid exam and final exam "

Rule: " ?\*a can be ?\*b " , [ ?a = ?b ]  
 Binding: {'a': ['marks', 'exams'], 'b': ['distributed', 'into', 'mid', 'exam', 'and', 'final', 'exam']}  
 Input: " marks exams "  
 Input: " distributed into mid exam and final exam "  
 Rule: " distributed into ?\*a and ?\*b " , [ ?a + ?b ]  
 Binding: {'a': ['mid', 'exam'], 'b': ['final', 'exam']}  
 Input: " mid exam "  
 Input: " final exam "  
 Output: ['mid', '+', 'final']  
 Output: ['marks', '=', ['mid', '+', 'final']]  
 Input: " internal marks can be divided into quizzes and assignments "  
 Rule: " ?\*a can be ?\*b " , [ ?a = ?b ]  
 Binding: {'a': ['internal', 'marks'], 'b': ['divided', 'into', 'quizzes', 'and', 'assignments']}  
 Input: " internal marks "  
 Input: " divided into quizzes and assignments "  
 Rule: " divided into ?\*a and ?\*b " , [ ?a + ?b ]  
 Binding: {'a': ['quizzes'], 'b': ['assignments']}  
 Input: " quizzes "  
 Input: " assignments "  
 Output: ['quizzes', '+', 'assignments']  
 Output: ['internal', '=', ['quizzes', '+', 'assignments']]  
 Input: " find total marks obtained by Asif he has obtained "  
 Rule: " find ?\*a " , [ To-find = ?a ]  
 Binding: {'a': ['total', 'marks', 'obtained', 'by', 'Asif', 'he', 'has', 'obtained']}  
 Input: " total marks obtained by Asif he has obtained "  
 Output: ['To-find', '=', 'total']  
 Input: " 8 marks in quizzes "  
 Rule: " ?\*a in ?\*b " , [ ?a = ?b ]  
 Binding: {'a': ['8', 'marks'], 'b': ['quizzes']}  
 Input: " 8 marks "  
 Input: " quizzes "  
 Output: ['8', '=', 'quizzes']  
 Input: " 6 marks in assignments "  
 Rule: " ?\*a in ?\*b " , [ ?a = ?b ]  
 Binding: {'a': ['6', 'marks'], 'b': ['assignments']}  
 Input: " 6 marks "  
 Input: " assignments "  
 Output: ['6', '=', 'assignments']  
 Input: " 10 marks in mid "  
 Rule: " ?\*a in ?\*b " , [ ?a = ?b ]  
 Binding: {'a': ['10', 'marks'], 'b': ['mid']}  
 Input: " 10 marks "  
 Input: " mid "  
 Output: ['10', '=', 'mid']  
 Input: " 20 marks in final exam "  
 Rule: " ?\*a in ?\*b " , [ ?a = ?b ]  
 Binding: {'a': ['20', 'marks'], 'b': ['final', 'exam']}  
 Input: " 20 marks "  
 Input: " final exam "  
 Output: ['20', '=', 'final']  
 Input: " 12 in practical lab work "

Rule: " ?\*a in ?\*b " , [ ?a = ?b ]

Binding: {'a': ['12'], 'b': ['practical', 'lab', 'work']}

Input: " 12 "

Input: " practical lab work "

Output: ['12', '=', 'practical']

Input: " "

UNSOLVED EQUATIONS

( theory + practical ) = total

theory = ( internal + marks )

marks = ( mid + final )

internal = ( quizzes + assignments )

To-find = total

8 = quizzes

6 = assignments

10 = mid

20 = final

12 = practical

## B. TRACING SOLVE EQUATIONS MODULE

| R.Call No | SOLVED EQUATIONS                                                                                                                                                                                                                                                                                                                                                                 | UN-SOLVED EQUATIONS    |
|-----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|
| 1         | ( theory + practical ) = total<br>theory = ( internal + marks )<br>marks = ( mid + final )<br>internal = ( quizzes + assignments )<br>To-find = total<br>8 = quizzes<br>6 = assignments<br>10 = mid<br>20 = final<br>12 = practical<br><br><b>1. Selection :</b> 8 = quizzes<br><b>2. Isolation :</b> quizzes = 8<br><b>3. Evaluate (RHS) :</b> quizzes = 8<br><b>4. Replace</b> | [ ]                    |
| 2         | ( theory + practical ) = total<br>theory = ( internal + marks )<br>marks = ( mid + final )<br>internal = ( <b>8</b> + assignments )<br>To-find = total<br>6 = assignments<br>10 = mid<br>20 = final<br>12 = practical<br><br><b>1. Selected:</b> 6 = assignments<br><b>2. Isolation:</b> assignments = 6<br><b>3. Evaluate (RHS) :</b> assignments = 6                           | [ quizzes = 8<br><br>] |

|          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                  |
|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|          | <b>4. Replace</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                  |
| <b>3</b> | <p> <math>(\text{theory} + \text{practical}) = \text{total}</math><br/> <math>\text{theory} = (\text{internal} + \text{marks})</math><br/> <math>\text{marks} = (\text{mid} + \text{final})</math><br/> <math>\text{internal} = (8 + 6)</math><br/> <math>\text{To-find} = \text{total}</math><br/> <math>10 = \text{mid}</math><br/> <math>20 = \text{final}</math><br/> <math>12 = \text{practical}</math> </p> <p> <b>1. Selected:</b> <math>\text{internal} = (8 + 6)</math><br/> <b>2. Isolation:</b> <math>\text{internal} = (8 + 6)</math><br/> <b>3. Evaluate (RHS) :</b> <math>\text{internal} = 14</math><br/> <b>4. Replace</b> </p> | <p> <math>[\text{quizzes} = 8]</math><br/> <math>\text{assignments} = 6</math> </p>                                                                                                              |
| <b>4</b> | <p> <math>(\text{theory} + \text{practical}) = \text{total}</math><br/> <math>\text{theory} = (14 + \text{marks})</math><br/> <math>\text{marks} = (\text{mid} + \text{final})</math><br/> <math>\text{To-find} = \text{total}</math><br/> <math>10 = \text{mid}</math><br/> <math>20 = \text{final}</math><br/> <math>12 = \text{practical}</math> </p> <p> <b>1. Selected:</b> <math>10 = \text{mid}</math><br/> <b>2. Isolation:</b> <math>\text{mid} = 10</math><br/> <b>3. Evaluate (RHS) :</b> <math>\text{mid} = 10</math><br/> <b>4. Replace</b> </p>                                                                                   | <p> <math>[\text{quizzes} = 8]</math><br/> <math>\text{assignments} = 6</math><br/> <math>\text{internal} = 14</math> </p>                                                                       |
| <b>5</b> | <p> <math>(\text{theory} + \text{practical}) = \text{total}</math><br/> <math>\text{theory} = (14 + \text{marks})</math><br/> <math>\text{marks} = (10 + \text{final})</math><br/> <math>\text{To-find} = \text{total}</math><br/> <math>20 = \text{final}</math><br/> <math>12 = \text{practical}</math> </p> <p> <b>1. Selected:</b> <math>20 = \text{final}</math><br/> <b>2. Isolation:</b> <math>\text{final} = 20</math><br/> <b>3. Evaluate (RHS) :</b> <math>\text{final} = 20</math><br/> <b>4. Replace</b> </p>                                                                                                                       | <p> <math>[\text{quizzes} = 8]</math><br/> <math>\text{assignments} = 6</math><br/> <math>\text{internal} = 14</math><br/> <math>\text{mid} = 10</math> </p>                                     |
| <b>6</b> | <p> <math>(\text{theory} + \text{practical}) = \text{total}</math><br/> <math>\text{theory} = (14 + \text{marks})</math><br/> <math>\text{marks} = (10 + 20)</math><br/> <math>\text{To-find} = \text{total}</math><br/> <math>12 = \text{practical}</math> </p> <p> <b>1. Selected:</b> <math>\text{marks} = (10 + 20)</math><br/> <b>2. Isolation:</b> <math>\text{marks} = (10 + 20)</math><br/> <b>3. Evaluate (RHS) :</b> <math>\text{marks} = 30</math><br/> <b>4. Replace</b> </p>                                                                                                                                                       | <p> <math>[\text{quizzes} = 8]</math><br/> <math>\text{assignments} = 6</math><br/> <math>\text{internal} = 14</math><br/> <math>\text{mid} = 10</math><br/> <math>\text{final} = 20</math> </p> |

|    |                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                           |
|----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7  | $(\text{theory} + \text{practical}) = \text{total}$<br>$\text{theory} = (14 + 30)$<br>$\text{To-find} = \text{total}$<br>$12 = \text{practical}$<br><br><b>1. Selected:</b> $\text{theory} = (14 + 30)$<br><b>2. Isolation:</b> $\text{theory} = (14 + 30)$<br><b>3. Evaluate (RHS) :</b> $\text{theory} = 44$<br><b>4. Replace</b> | [<br>quizzes = 8<br>assignments = 6<br>internal = 14<br>mid = 10<br>final = 20<br>marks = 30<br>]                                                                         |
| 8  | $(44 + \text{practical}) = \text{total}$<br>$\text{To-find} = \text{total}$<br>$12 = \text{practical}$<br><br><b>1. Selected:</b> $12 = \text{practical}$<br><b>2. Isolation:</b> $\text{practical} = 12$<br><b>3. Evaluate (RHS) :</b> $\text{practical} = 12$<br><b>4. Replace</b>                                                | [<br>quizzes = 8<br>assignments = 6<br>internal = 14<br>mid = 10<br>final = 20<br>marks = 30<br>theory = 44<br>]                                                          |
| 9  | $(44 + 12) = \text{total}$<br>$\text{To-find} = \text{total}$<br><br><b>1. Selected:</b> $(44 + 12) = \text{total}$<br><b>2. Isolation:</b> $\text{total} = (44 + 12)$<br><b>3. Evaluate (RHS) :</b> $\text{total} = 56$<br><b>4. Replace</b>                                                                                       | [<br>quizzes = 8<br>assignments = 6<br>internal = 14<br>mid = 10<br>final = 20<br>marks = 30<br>theory = 44<br>practical = 12<br>]                                        |
| 10 | $\text{To-find} = 56$<br><br><b>1. Selected:</b> $\text{To-find} = 56$<br><b>2. Isolation:</b> $\text{To-find} = 56$<br><b>3. Evaluate (RHS) :</b> $\text{To-find} = 56$<br><b>4. Replace</b>                                                                                                                                       | [<br>quizzes = 8<br>assignments = 6<br>internal = 14<br>mid = 10<br>final = 20<br>marks = 30<br>theory = 44<br>practical = 12<br>total = 56<br>]                          |
| 10 | [ ]                                                                                                                                                                                                                                                                                                                                 | [<br>quizzes = 8<br>assignments = 6<br>internal = 14<br>mid = 10<br>final = 20<br>marks = 30<br>theory = 44<br>practical = 12<br>total = 56<br>$\text{To-find} = 56$<br>] |



## STUDENT COMPLETE CODE

```

from patmatch import * # or write the patmatch code in this file
operators = "+-*/%"
numbers = "0123456789"
letters = "abcdefghijklmnopqrstuvwxyzABCDEFGHIJKLMNOPQRSTUVWXYZ"
student_rules = [
    ('find ?*a' , ['To-find' , '=', '?a']),
    ('?*a is equal to ?*b' , ['?a' , '=', '?b']),
    ('?*a makes ?*b' , ['?a' , '=', '?b']),
    ('?*a are calculated by ?*b' , ['?a' , '=', '?b']),
    ('?*a calculated by ?*b' , ['?a' , '=', '?b']),
    ('?*a can be calculated as ?*b' , ['?a' , '=', '?b']),
    ('?*a calculated as ?*b' , ['?a' , '=', '?b']),
    ('?*a can be ?*b' , ['?a' , '=', '?b']),
    ('?*a equal to ?*b' , ['?a' , '=', '?b']),
    ('?*a equals ?*b' , ['?a' , '=', '?b']),
    ('?*a is ?*b' , ['?a' , '=', '?b']),
    ('?*a are ?*b' , ['?a' , '=', '?b']),
    ('?*a in ?*b' , ['?a' , '=', '?b']),
    ('?*a plus ?*b' , ['?a' , '+', '?b']),
    ('?*a minus ?*b' , ['?a' , '-', '?b']),
    ('?*a divided by ?*b' , ['?a' , '/', '?b']),
    ('twice ?*a' , ['2' , '*', '?a']),
    ('sum ?*a and ?*b' , ['?a' , '+', '?b']),
    ('adding ?*a and ?*b' , ['?a' , '+', '?b']),
    ('distributed into ?*a and ?*b' , ['?a' , '+', '?b']),
    ('divided into ?*a and ?*b' , ['?a' , '+', '?b']),
    ('one half ?*a' , ['?a' , '/', '2']),
    ('half ?*a' , ['?a' , '/', '2']),
    ('square ?*a' , ['?a' , '*', '?a']),
    ('?*a times ?*b' , ['?a' , '*', '?b']),
    ('?*a multiplied by ?*b' , ['?a' , '*', '?b']),
    ('?*a multiplied with ?*b' , ['?a' , '*', '?b']),
    ('difference between ?*a and ?*b' , ['?b' , '-', '?a']),
    ('?*a % less than ?*b' , ['?b' , '/', ['1' , '-', ['?a' , '/', '100']]])
]

def toeqstr(elst):
    ans = ''
    if type(elst) is str:
        return elst
    for item in elast:
        if type(item) is list:
            ans = ans + " ( " + toeqstr(item) + " )"
        else:
            if ans == '':
                ans = item
            else:
                ans = ans + ' ' + str( item)
    return ans

def replace_item(n,o,lst):
    if lst == []:
        return []
    elif o == lst:
        return n
    elif type(lst) is not list:
        return lst
    else:
        ans = replace_item(n,o,lst[1:])
        ans.insert(0,replace_item(n,o,lst[0]))
        return ans

```

```
def translate_to_exp(eqwlst):
    print('Input: "', toeqstr(eqwlst), '"')
    for pattern, response in student_rules:
        pattern = pattern.split()
        bindings = match_pattern(pattern, eqwlst)
        if bindings:
            print('Rule:', toeqstr(pattern), ", ['", toeqstr(response), ']')
            print('Binding:', bindings)
            for var, val in bindings.items():
                #print(var)
                tval = translate_to_exp(val)
                response = replace_item(tval, '?' + var, response)
            print('Output:', response)
            return response
    if eqwlst == []:
        return eqwlst
    print('Output:', '"', eqwlst[0], '"')
    return eqwlst[0]

def remove_noise(txt):
    nwords = ['a', 'the', 'number', 'of', 'this', 'if', 'that',
              '$', 'and', 'his', 'while']

    for w in nwords:
        if txt.count(w) > 0:
            txt = [a for a in txt if a != w]
    return txt

def student():
    userinput = input("Enter Statement:")
    userinput = userinput.lower()
    engEqs = userinput.split('.')
    eqs = []
    for engeq in engEqs:
        engeq = engeq.split()
        eqwlst = remove_noise(engeq)
        eq1 = translate_to_exp(eqwlst)
        eqs.append(eq1)
    eqs = [eq for eq in eqs if eq != []]
    solve_equations(eqs)

def solve_equations(eqs):
    print("UNSOLVED EQUATIONS")
    print_equations(eqs)
    print('SOLVED EQUATIONS')
    solved = solve(eqs)
    print_equations(solved)

def print_equations(eqs):
    for eq in eqs:
        print(toeqstr(eq))
```

```
def solve(eqs, solved=[]):
    if len(eqs) == 0:
        return solved
    ans = find_one_unknowneq(eqs)
    var = ans[0]
    eq = ans[1]
    if eq != [] and var is not None:
        sol=solve_equation(var,eq)
        value = sol[1]
        eqs.remove(eq)
        eqs = replace_item(value, var,eqs)
        solved.append([var, '=', value])
        temp= solve(eqs,solved)
        return temp
    return solved

def solve_equation(var, eq):
    eq2 = isolate(var, eq)
    return eq2[0], str(eval(toeqstr(eq2[2])))

def find_one_unknowneq(eqs):
    for eq in eqs:
        ans = one_unknown2(eq)
        count = ans[1]
        var = ans[0]
        if count == 1:
            return var,eq
    return None, []

def one_unknown2(equation):
    c = 0
    var = None
    for item in equation:
        if type(item) is list:
            ans= one_unknown2(item)
            c += ans[1]
            if ans[1] != 0:
                var = ans[0]
        else:
            flag = False
            for ch in item:
                if ch in letters:
                    flag = True
            if flag == True:
                c += 1
                var = item
    return var,c

def in_exp(itm, lst):
    if type(lst) is str:
        return itm == lst
    elif lst == []:
        return False
    elif type(lst) is list:
        return in_exp(itm, lst[0]) or in_exp(itm, lst[1:])
```

```
def reverse_op(op):
    if op == '+':
        return '-'
    elif op == '-':
        return '+'
    elif op == '*':
        return '/'
    elif op == '/':
        return '*'
    elif op == '=':
        return '='

def isolate(var, eq):
    lhs= eq[0]
    op = eq[1]
    rhs= eq[2]
    if var == lhs: #CASE 1: X = A
        return eq
    elif var == rhs:
        return [rhs, op, lhs]
    elif type(lhs) is str:
        return isolate(var, [rhs,op,lhs])
    elif type(rhs) is list and op == '=' and in_exp(var,rhs):
        return isolate(var, [rhs,op,lhs])
    elif in_exp(var,lhs[0]):
        return isolate(var, [lhs[0], '=', [rhs, reverse_op(lhs[1]), lhs[2]]])
    elif lhs[1] == '+' or lhs[1] == '*':
        return isolate(var, [lhs[2], '=', [rhs, reverse_op(lhs[1]), lhs[0]]])
    else:
        return isolate(var, [lhs[2], '=', [lhs[0], lhs[1], rhs]])

student()
```